INCH - POUND

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DETAIL SPECIFICATION

STRONTIUM PEROXIDE

Reactivated after 24 September 2024 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and test for strontium peroxide.

1.2 Classification. Strontium peroxide is of the following grades, as specified (see 6.2).

TABLE I. Classification.

Grade	Description
Grade B	No hygroscopicity
Grade C	0.25 percent (%) hygroscopicity

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/)

2.3 Non-government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Woven Test Sieve Cloth and Test Sieves
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM D1193	-	Standard Specification for Reagent Water

(Copies of these documents are available from https://www.astm.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material.

3.2.1 Grade B. Strontium peroxide shall be manufactured by the wet precipitation method and shall consist of anhydrous strontium peroxide.

3.2.2 Grade C. Strontium peroxide shall be manufactured by the wet precipitation method and shall consist of anhydrous strontium peroxide of a definite hygroscopicity requirement to control the stability of the material.

3.3 Physical and chemical requirements. Strontium peroxide shall be in accordance with Table II.

TABLE II. Physical and chemical requirements for strontium peroxide.

Properties	Characteristics	
	Grade B	Grade C
Available oxygen, weight percent minimum (wt% min)	12.3	12.3
Moisture, weight percent maximum (wt% max)	------	0.1
Hygroscopicity, wt% max	------	0.25
Apparent density, grams per milliliter minimum, (g/mL min)	1.30-2.00	0.85
Barium, wt% max	1.3	0.45
Calcium, wt% max	0.3	0.15
Magnesium, wt% max	0.05	0.02
Sodium, wt% max	0.1	0.05
Iron, wt% max	0.09	0.03
Aluminum, wt% max	0.1	0.05
Nitrate, calculated as strontium nitrate ($(SrNO3)_2$), wt% max	0.1	0.05
Chloride, calculate as strontium chloride ($SrCl_2$), wt% max	0.1	0.10

3.4 Granulation. Each grade of strontium peroxide shall be in accordance with the following Table III.

TABLE III. Granulation.

Through US Alternative sieve No.	Grade B	Grade C
80, percent, min,	------	100
100, percent, min,	98	98
140, percent, min,	80	80

3.5 Workmanship. Strontium peroxide shall be a free flowing white powder, free from lumps, grit, visible impurities, and foreign material such as dirt, rust, paint, wood.

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4. VERIFICATION

TABLE IV. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.3	Available oxygen	X		X	X	X	X	4.4.2
3.3	Moisture	X		X	X	X	X	4.4.3
3.3	Hygroscopicity	X		X	X	X	X	4.4.4
3.3	Apparent density	X		X	X	X	X	4.4.5
3.3	Barium, calcium, magnesium, iron, aluminum, and sodium content	X		X	X	X	X	4.4.6 to 4.4.9
3.3	Nitrate as $(SrNO_3)_2$ content	X		X	X	X	X	4.4.10
3.3	Chloride as ($SrCl_2$)	X		X	X	X	X	4.4.11
3.4	Granulation	X		X	X	X	X	4.4.12
3.5	Workmanship	X		X	X	X	X	4.4.13

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table V. Samples of not less than one (1) kg of strontium peroxide collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE V. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method
Critical	None defined		
Major			
101	Available Oxygen	3.3	4.4.2
102	Moisture	3.3	4.4.3
103	Hygroscopicity	3.3	4.4.4
104	Apparent density	3.3	4.4.5
105	Barium, calcium, magnesium, iron, aluminum, and sodium content	3.3	4.4.6 to 4.4.9
108	Nitrate as $(SrNO_3)_2$	3.3	4.4.10
109	Chloride as ($SrCl_2$)	3.3	4.4.11
110	Granulation	3.4	4.4.12
Minor			
201	Workmanship	3.5	4.4.13

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements, the first article shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of a quantity of strontium peroxide that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of strontium peroxide produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Strontium peroxide sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the strontium peroxide is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than five hundred (500) grams shall be formed which is assumed to be representative from a batch. All composite samples shall be verified in accordance with 4.3.3. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Strontium peroxide batches shipped in containers (e.g., drums, totes, boxes, etc.) shall be sampled using Verification Level I of MIL-STD-1916, where one unit is defined as one container. Each container shall be sampled in accordance with ASTM E300 using a sample thief. Approximately equal portions shall be taken from each selected container. This material shall be placed in a secondary container and mixed to form a

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composite sample consisting of a minimum of five hundred (500) grams. The composite sample shall be tested and verified in accordance with 4.3.3. If any sample fails to comply with any applicable test requirements the lot represented by the sample shall be rejected.

4.3.3 Strontium peroxide.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Available oxygen	4.3.2	3.3	4.4.2
102	Moisture	4.3.2	3.3	4.4.3
103	Hygroscopicity	4.3.2	3.3	4.4.4
104	Apparent density	4.3.2	3.3	4.4.5
105	Barium, calcium, magnesium, iron, aluminum, and sodium content	4.3.2	3.3	4.4.6 to 4.4.9
106	Nitrate as $(SrNO_3)_2$	4.3.2	3.3	4.4.10
107	Chloride as ($SrCl_2$)	4.3.2	3.3	4.4.11
108	Granulation	4.3.2	3.4	4.4.12
Minor				
201	Workmanship	4.3.2	3.5	4.4.13

4.4 Test methods. The methods contained herein are recommended based on prior experience. However, deviations may be permitted, provided they are justified and based on need for safety, speed, convenience, equipment availability, or cost, and do not adversely affect the accuracy and precision of the method. The operator shall note any deviations used. (See 6.8).

4.4.1 General. Unless indicated otherwise, distilled water, in accordance with ASTM D1193 Type III and analytical grade reagents, shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents used in the test, and corrections shall be applied when necessary.

4.4.2 Available oxygen.

4.4.2.1 Acid reagent. Add 100 milliliters (mL) of concentrated perchloric acid, 30 mL of concentrated phosphoric acid, and 0.1 grams (g) of manganese chloride ($MnCl_2 \cdot 4H_2O$) to water and dilute to 1 liter (L).

4.4.2.2 Procedure. Weigh approximately 0.35 to 0.40 g of the sample in a 300 mL beaker. Record the sample weight to the nearest 0.0001 g. Wash down the sides of the beaker with 70 mL of acid reagent. From a 100 mL burette, run into the beaker, about ¾ of the amount of 0.1N potassium permanganate solution needed for the titration (see 6.6). Stir the solution in the beaker mechanically with a magnetic stir bar. Cover the beaker with a watch glass or plastic film containing a hole for the tip of the burette and continue stirring. After solution of the sample is

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complete, titrate with the 0.1N potassium permanganate solution to a pink color that persists for 1 minute. Calculate the available oxygen percent as follows:

$$\text{Available oxygen percent} = \frac{0.8 \times V \times N}{W}$$

Where: V = Volume of potassium permanganate solution, mL
N = Normality of potassium permanganate solution
W= Weight of sample, g

NOTE: The above directions shall be rigidly followed. The sample should be brought completely in solution as quickly as possible and then titrated immediately. Any delay in the titration will cause low results, regardless of stabilizer or lowering of the temperature. The solutions shall not be greater than room temperature (too high temperature causes decomposition), nor less than room temperature (too low temperature retards dissolution of the sample). Room temperature is considered 18°C to 24°C.

4.4.3 Moisture. Transfer a 6.00 ± 0.05 g sample to a tared weighing dish (approximately 6 cm diameter and 2.5 cm deep), weigh and record the gross weight to the nearest 0.001 g. Dry the sample at 130º ± 3 ºC for 3 hours, cool in a desiccator, weigh and record the measurement to the nearest 0.001 g. Calculate the moisture percent as follows:

$$\text{Moisture percent} = \frac{100 \times A}{W}$$

Where: A = Loss in weight, g
W = Weight of sample, g

Save the dried samples for the hygroscopicity determination (see 4.4.4).

4.4.4 Hygroscopicity. Place the dried sample from the moisture determination into a desiccator, 10 L capacity, containing 1 L of a sulfuric acid solution (30.1 ± 0.2% sulfuric acid by weight) for producing 75% relative humidity at 25ºC. Allow the dishes to remain in the desiccator for 48 hours and weigh again. Place the dishes back into the desiccator for 24 hours and weigh again. Repeat for 24-hour intervals until constant weight within one (1) milligram is obtained. Usually, the weight will not change significantly after the initial 48-hour period. Calculate the hygroscopicity percent as follows:

$$\text{Hygroscopicity percent} = \frac{100 \times G}{W}$$

Where: G = Gain in weight, g
W = Weight of sample, g

4.4.5 Apparent density. Weigh a 50 mL graduated cylinder (ground-glass top) into which has been inserted a rubber stopper. Remove the stopper and place a powder funnel on the top of the cylinder to transfer approximately 30.0 g of sample. With the stopper inserted, drop the

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cylinder vertically 30 times from a height of 2.0 inches (in) onto a hard leather pad (or book), regulating the height of fall by inserting the cylinder through a support ring of suitable size. Tap the sides of the cylinder gently with a straight edge to level the height, read the volume occupied, and weigh. Calculate the apparent density as follows:

$$\text{Apparent density} = \frac{W}{V}$$

Where: W = Weight of sample, g
V = Volume occupied, mL

4.4.6 Aluminum, barium, calcium, iron, magnesium, and sodium by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES) (preferred method).

4.4.6.1 Equipment and supplies.

a. Inductively coupled, argon-plasma, optical emission spectrometer, dual view
b. Compressed argon, as per ICP-OES manufacturer recommendations
c. Class A volumetric flasks and pipettes
d. Class A or B plastic volumetric flasks, 100 mL capacity
e. 150 mL plastic beakers, capable of withstanding 100oC and action of dilute mineral acids.
f. Adjustable volume pipettes with calibration verified at most a month prior: 0.1 - 1 mL and 1 - 5 mL.
g. Analytical balance with 200 g capacity and 0.1 mg resolution.

4.4.6.2 Stock solutions and reagents.

4.4.6.2.1 Reagents. All chemicals used shall be reagent or ACS grade, or as specified below.

a. Water, deionized (DI) or distilled, conforming to ASTM D1193 Type I or II; all subsequent mentions of water in section 4.4.6 refer hereto.
b. Nitric Acid (HNO_3), concentrated, 67-70 wt %, trace-metal grade.
c. Nitric acid 1:10 v/v: dilute 100 mL of trace-metal grade concentrated nitric acid to 1 L with water.
d. Nitric acid 1:1 v/v: dilute 500 mL of trace-metal grade concentrated nitric acid to 1 L with water.
e. Hydrochloric Acid (HCl), concentrated, 34-37 wt %, trace-metal grade.

4.4.6.2.2 Stock solutions.

4.4.6.2.2.1 Stock solutions (preferred). If available, commercially available ICP stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology, or international measurement standards) shall be used. Stock solutions shall be stored and replaced in accordance with manufacturer recommendations. Common, commercially available stock solutions are as follows:

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a. Certified Al stock solution, 1000 ± 5 mg/L in 5-10% (v/v) HNO_3 or HCl
b. Certified Ba stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl
c. Certified Ca stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl
d. Certified Fe stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl
e. Certified Mg stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl
f. Certified Na stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl
g. Certified Y stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl

Note: The maximum recommended shelf life of stock solutions is one (1) year from date of initial preparation when kept tightly sealed at 5-25°C.

4.4.6.2.2.1 Stock solutions (alternate). Stock solutions shall be prepared in-house from high purity metals or their salts. All the materials used to prepare the stock solutions shall be 99.9% purity or better.

4.4.6.2.2.1.1 Stock aluminum (Al) solution preparation (aluminum 1000 mg/L). Place 1.000 ± 0.005 g of elemental aluminum in 200 mL of 1:1 nitric acid then slowly add 20 mL of hydrochloric acid, agitate until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

4.4.6.2.2.1.2 Stock Barium (Ba) solution preparation (barium 1000 mg/L). Dissolve 1.903 ± 0.005 g of barium nitrate in 1:10 nitric acid, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with 1:10 nitric acid.

4.4.6.2.2.1.3 Stock Calcium (Ca) solution preparation (calcium 1000 mg/L). Dissolve 2.497 ± 0.005 g of calcium carbonate in 1:10 nitric acid, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with 1:10 nitric acid.

4.4.6.2.2.1.4 Stock Iron (Fe) solution preparation (iron 1000 mg/L). Weigh 1.000 ± 0.005 g of elemental iron into 50 mL water, then slowly add 200 mL of 1:1 nitric acid, agitate until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

4.4.6.2.2.1.5 Stock Magnesium (Mg) solution preparation (magnesium 1000 mg/L). Weigh 1.000 ± 0.005 g of elemental magnesium, add approx. 50 mL water, then slowly add 200 mL of 1:1 nitric acid, agitate until fully dissolved (heat if necessary), transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

4.4.6.2.2.1.6 Stock Sodium (Na) solution preparation (sodium 1000 mg/L). Dissolve 2.305 ± 0.005 g of sodium carbonate in 1:10 nitric acid, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with 1:10 nitric acid.

4.4.6.2.2.1.7 Stock Yttrium (Y) solution preparation (yttrium 1000 mg/L). Weigh 1.000 ± 0.005 g of elemental yttrium, add approx. 50 mL water, then slowly add 200 mL of 1:1 nitric acid, agitate until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

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4.4.6.3 Preparation of samples and reagent blank. The calibration and analysis shall be performed using the method of standard additions (MSA). Additionally, yttrium is used as internal standard to control unwanted variation. Each sample shall be prepared and analyzed using six (6) replicates, 1.000 g each, per Tables VI & VII. The reagent blank shall be carried through the entire procedure and its reading shall be subtracted from all samples.

4.4.6.3.1 Procedure. Weigh 1.000 ± 0.002 g of sample into each of six (6) 150-mL beakers. To three of the beakers add the amounts of stock solutions indicated in Table VI. Note: Yttrium stock solution is added separately from all others, after the sample solution has been heated and cooled, and just before transfer to plastic 100 mL volumetric flask. Add 15 mL of DI water, then slowly add 10 mL of nitric acid, mix well, and heat the solutions briefly over a hotplate (100°C surface temperature) for about 10 min to drive oxygen off. The solutions should become colorless without any precipitation. Cool the solutions, add 0.200 mL of 1000 mg/L yttrium stock, and transfer quantitatively to 100 mL volumetric flasks. Make up to volume with DI water and mix well.

TABLE VI. Volumes (mL) of each *1000 mg/L stock indicated for sample preparation.

Sample identifier	Weight, g	Al	Ba	Ca	Fe	Mg	Na	Y**
Reagent blank	0	0	0	0	0	0	0	0.200
Calibration spike 0 (replicate 1)	1.000 ± 0.002	0	0	0	0	0	0	0.200
Calibration spike 0 (replicate 2)	1.000 ± 0.002	0	0	0	0	0	0	0.200
Calibration spike 0 (replicate 3)	1.000 ± 0.002	0	0	0	0	0	0	0.200
Calibration spike 1	1.000 ± 0.002	0.50	4.50	1.50	0.30	0.20	0.50	0.200
Calibration spike 2	1.000 ± 0.002	1.50	15.0	3.50	1.00	0.50	1.50	0.200
Calibration spike 3	1.000 ± 0.002	2.50	30.0	7.50	2.00	1.20	2.50	0.200

Remarks:
* Actual concentrations of commercially obtained stock solutions, or of those prepared in-house, shall be used. Added concentrations in Table VII shall be adjusted proportionally.
**Yttrium stock solution is added after solution has cooled and just before transferring to 100 mL flask.

TABLE VII. MSA: nominal added concentrations (mg/L) when sample prepared per Table VI.

Sample identifier	Al	Ba	Ca	Fe	Mg	Na	Y
Reagent blank	0	0	0	0	0	0	2
Calibration spike 0 (replicate 1)	0	0	0	0	0	0	2
Calibration spike 0 (replicate 2)	0	0	0	0	0	0	2
Calibration spike 0 (replicate 3)	0	0	0	0	0	0	2
Calibration spike 1	5	45	15	3	2	5	2
Calibration spike 2	15	150	35	10	5	15	2
Calibration spike 3	25	300	75	20	12	25	2

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4.4.6.4 Calibration and quantitation (recommended). The wavelengths and view modes shown in Table VIII are suitable for a ThermoFisher iCAP6500 Duo ICP system with iTeva version 2.5 software. Additional method development may be needed for other systems. Prepare the instrument in accordance with manufacturer recommendations. Use the wavelengths found in Table VIII for analysis of the various elements.

TABLE VIII. Suggested wavelengths and view modes for ICP-OES analysis.

Element	Wavelength*, nm	View	Referencing elements
Al	396.1, 309.2, 394.4	axial	----
Ba	413.0, 416.6, 234.7	radial	----
Ca	317.9, 315.8, 373.6	radial	----
Fe	259.9, 238.2, 259.8	axial	----
Mg	279.5, 280.2, 285.2	radial	----
Na	588.9, 589.5	radial	
Y (internal standard)	371.0	axial	Al, Fe
Y (internal standard)	371.0	radial	Ba, Ca, Mg, Na

Note:
* Wavelengths are shown from left to right in order of decreasing preference.

Note: Due to the high solids content and low analyte concentrations, rinse the sample introduction system with 1:10 nitric acid before the analysis and between samples.

4.4.6.4.1 Procedure. Perform calibration in standard additions mode using the following sampling order: reagent blank → calibration spike 0 (replicate 1) → calibration spike 1→ calibration spike 2 → calibration spike 3. The generated calibration shall be linear with $R^2 \geq 0.999$. The software will calculate and report analyte concentrations in calibration spike 0 (rep 1). Alternately, or as a check, the concentration may be calculated outside the software, using commonly accepted evaluation of MSA. Repeat the sampling sequence for replicates 2 and 3 of calibration spike 0, and obtain another two sets (total of three) of analyte concentrations. The maximum and minimum results for the three replicates of calibration spike 0 shall differ by no more than the values indicated in Table IX.

TABLE IX. Acceptable range (maximum minus minimum) for the three replicates of calibration spike 0, all values in mg/L.

Element	Grade B	Grade C
Al	0.5	0.5
Ba	5	2
Ca	1	0.5
Fe	0.5	0.5
Mg	0.5	0.5
Na	0.5	0.5

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Note: If the above condition is met, calculate the average concentration of each analyte in calibration spike 0 based on the three replicates.

Calculate and report the weight percent of each elemental impurity as follows:

$$\text{Weight percent of the elemental impurity} = 0.01 \times C$$

Where: C = Average analyte concentration in calibration spike 0, mg/L

4.4.7 Barium, calcium, magnesium, and sodium (alternate method) (see 6.5).

4.4.7.1 Apparatus. Atomic absorption spectrophotometer.

4.4.7.2 Reagents.

4.4.7.2.1 Standard barium (Ba) solution preparation (1 mL = 1.0 mg Ba). Dissolve 1.7790 g of barium chloride ($BaCl_2 \cdot 2H_2O$) in water and dilute to 1 L in a volumetric flask.

4.4.7.2.2 Standard calcium (Ca) solution preparation (1 mL= 0.5 mg Ca). Transfer 1.2490 g of calcium carbonate to a 400 mL beaker and add 100 mL of water. Slowly add 5 mL of hydrochloric acid, heat to boiling point, boil for 1 to 2 minutes, cool to room temperature, and dilute to 1 L in a volumetric flask.

4.4.7.2.3 Standard magnesium (Mg) solution preparation (1 mL = 0.05 mg Mg). Transfer 0.0500 g of magnesium ribbon to a 250 mL beaker and add 50 mL of water. Slowly add 5 mL of hydrochloric acid, heat to boiling point, boil for 1 or 2 minutes, cool to room temperature, and dilute to 1 L in a volumetric flask.

4.4.7.2.4 Standard sodium (Na) solution preparation (1 ml = 0.05 mg Na). Dissolve 0.1271 g of sodium chloride in water and dilute to 1 L in a volumetric flask.

4.4.7.3 Procedure. Transfer a 1.0 g sample to a 250 mL beaker, add 30 mL of nitric acid (1 to 9), warm to dissolve, and then boil down to a volume of about 10 mL. Cool to room temperature and dilute to 100 mL in a volumetric flask. Dilute appropriate aliquots of the standard solutions of barium, calcium, magnesium, and sodium to 100 mL in volumetric flasks and measure the absorbance of the standards and the samples with the atomic absorption spectrophotometer in accordance with the manufacturer's recommendations. Calculate the weight in mg of barium, calcium, magnesium, and sodium in the sample (per 100 mL) and calculate the barium, calcium, magnesium, and sodium percent as follows:

$$\text{Barium, calcium, magnesium and sodium percent} = \frac{M}{10 \times W}$$

Where: M = Weight of barium, calcium, magnesium, or sodium (per 100 mL), mg
W = Weight of sample, g

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4.4.8 Iron (see 6.4).

4.4.8.1 Reagents.

4.4.8.1.1 Standard iron (Fe) solution preparation. (1 mL = 0.1 mg Fe). Transfer 0.1000 g of pure iron to a 250 mL beaker, add 50 mL of hydrochloric acid and warm until dissolved. Add 3 mL of hydrogen peroxide (30%), boil for 10 minutes, cool to room temperature, and dilute to 1 L in a volumetric flask.

4.4.8.1.2 Hydroxylamine hydrochloride ($NH_2OH•HCl$) solution preparation (5%). Dissolve 25.0 g of hydroxylamine hydrochloride in water and dilute to 500 mL.

4.4.8.1.3 Ortho-phenanthroline solution preparation (0.2%). Dissolve 1.0 g of ortho-phenanthroline in warm water, cool to room temperature and dilute to 500 mL.

4.4.8.1.4 Sodium acetate ($CH_3COON_a•3H_3O$) solution preparation (50%). Dissolve 250 g of sodium acetate in water and dilute to 500 mL.

4.4.8.2 Preparation of calibration. Accurately transfer 0.5, 1.0, 2.0, 2.5 and 3.0-mL potions of standard iron solution to 100 mL volumetric flasks. Carry along a reagent blank. Add 50 mL of water and 5 mL of hydroxylamine hydrochloride solution (5%) and allow to stand 15 minutes. Add 5 mL of ortho-phenanthroline solution (0.2%) and 10 mL of sodium acetate solution (50%) and dilute to the mark. Allow to stand 30 minutes or more and measure the absorbance at 500 nm with a spectrophotometer or colorimeter that has been set to 100% transmittance with the reagent blank. Plot the weight in mg of iron against the absorbance.

4.4.8.3 Procedure. Weigh in a 1.0 g sample into a 400 mL beaker, add 50 mL of water and 5 mL of hydrochloric acid, and evaporate to incipient dryness on the hot plate at low heat. Carry along a reagent blank. Add 50 mL of water and warm to dissolve the salts. If the solution is not clear, filter through a Whatman No. 42 filter paper, and wash with water. Wash into a 100 mL volumetric flask. Add 5 mL of hydroxylamine hydrochloride solution (5%) to the solution in the 100 mL volumetric flask and allow to stand 15 minutes. Add 5 mL of ortho-phenanthroline solution (0.2%) and 10 mL of sodium acetate solution (50%) and dilute to the mark. Allow to stand 30 minutes or more and measure the absorbance at 500 nm with a spectrophotometer or colorimeter that has been set to 100% transmittance with the reagent blank. Convert the reading to mg of iron by referring to the calibration curve. Calculate the iron percent as follows:

$$\text{Iron percent} = \frac{M}{10 \times W}$$

Where: M = Weight of iron as read from calibration curve, mg
W = Weight of sample on which color was developed, g.

4.4.9 Aluminum (see 6.4). Weigh approximately 5.0 g sample into a 250 mL beaker, add 100 mL of deionized (DI) water and 40 mL of hydrochloric acid, and heat on a hot plate until dissolved. Filter through a Whatman No. 42 filter paper into a 400 mL beaker and wash with DI water. Discard the precipitate. Add 4.0 g of ammonium chloride to the filtrate, cover the beaker

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with a watch glass and boil for 5 minutes to destroy the peroxide. Add a few drops of methyl red and concentrated ammonium hydroxide (approximately 30 mL) until the solution turns yellow. Allow the solution to stay hot for a few minutes. Filter through a Whatman No. 40 filter paper, and wash with hot 1% ammonium chloride that has been made ammoniacal to methyl red. Dissolve the precipitate from the filter paper into the original 250 mL beaker with about 20 mL of hydrochloric acid (1 to 1) and rinse with a bit of water (the filter may disintegrate in the process). Neutralize to yellow coloration of methyl red with concentrated ammonium hydroxide. Keep the solution hot for approximately 30 minutes to coagulate any precipitates. Filter through Whatman No.40, wash beaker and filter with hot ammoniacal 1% ammonium chloride. Place the filter with precipitate into tared platinum crucible, and dry, char and burn off paper in muffle furnace at approximately 400°C for 2 hours. Ignite crucible at 600°C for about 1 hour, and cool to room temperature. Add 3 mL of concentrated hydrofluoric acid and 2 drops of sulfuric acid and evaporate to near dryness. Ignite at 600°C for 30 to 60 minutes, cool to room temperature, and weigh the combined oxides of aluminum and iron. Calculate the percent aluminum as follows:

$$\text{Aluminum percent} = \frac{52.9 \times (A - (0.0143 \times B \times W))}{W}$$

Where: A = Weight of precipitate, g
B = Iron percent (as determined in 4.4.6 or 4.4.8)
W= Weight of sample, g

4.4.10 Nitrate. Weigh a 10.00 ± 0.05 g sample into a 500 mL Kjeldahl flask and dissolve in premixed solution containing 150 mL deionized water and 15 mL glacial acetic acid. Carry along a reagent blank. Swirl the flask until dissolved (approximately 20 mins). Cautiously add 15 mL 50 % wt. sodium hydroxide solution. Ensure the solution pH is at least 12 to universal pH paper, if not add a few more mL of base and re-check. Add approximately 1.0 g of finely ground Devarda's alloy, shake to thoroughly mix with flask contents, and immediately connect to a trap and a water-cooled condenser, the end of which dips below the surface of 50 mL of 4% boric acid solution contained in a 250 mL Erlenmeyer flask or other vessel which can then be sealed (extend the condenser tip with fluoropolymer tubing if needed). Warm the Kjeldahl flask at approximately 80ºC for 2 hours, then heat to the boiling point and distill to a total volume of 150 mL in the receiver. Remove the receiver immediately to prevent back-flow of distillate. Add 4 drops of 0.1% methyl purple indicator solution to the distillate (bromophenol blue or methyl red may be used as alternates) and titrate to the same shade of purple as the blank with 0.01N sulfuric acid; alternately use a pH meter to titrate to the same pH as the blank. Note: the blank shall not be titrated, but merely used as a color or pH reference. Calculate the nitrate percent (as strontium nitrate) as follows:

$$\text{Percent nitrate calculated as strontium nitrate} = \frac{10.58 \times V \times N}{W}$$

Where: V = volume of sulfuric acid used, mL
N = normality of sulfuric acid
W= weight of sample, g

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4.4.11 Chloride.

4.4.11.1 Chloride back-titration with thiocyanate (preferred method).

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4.4.11.1.1 Equipment and supplies.

a. 250-mL conical flasks
b. pH test paper, universal
c. Hotplate
d. Class A burette, 10 mL
e. Magnetic stirrer
f. Reagents
g. All chemicals used shall be reagent grade, or as specified below:
h. Water, deionized (DIW) or distilled, conforming to ASTM D1193 Type I or II
i. Nitric Acid (HNO3), concentrated, 67-70 wt%
j. Ferric nitrate, 10 wt%: dissolve 10 g of ferric nitrate nonahydrate in water and dilute to 100 mL. Acidify with concentrated nitric acid to pH 1.
k. Silver nitrate, 0.05N
l. Ammonium thiocyanate, 0.025N

4.4.11.1.2 Silver nitrate titer of thiocyanate. Place 4.00 mL of 0.05 N silver nitrate in a 250 mL flask and dilute to 100 mL with water. Add 5 mL of 10% ferric nitrate and mix well for 1 min. Titrate the sample solution with 0.025 N ammonium thiocyanate to a light brown color that persists for at least 15 sec. Record the volume of thiocyanate used. Calculate the silver nitrate titer of the thiocyanate as follows:

$$\text{Titer, (volume 0.05N silver nitrate per volume 0.025N ammonium thiocyanate)} = \frac{4}{V}$$

Where: V = volume of ammonium thiocyanate used, mL

4.4.11.1.3 Analysis of strontium peroxide for chloride. Each sample is analyzed using two (2) 5-gram replicates. Weigh 5.000 ± 0.050 g of strontium peroxide sample into a 250 mL conical flask, add a pre-mixed solution of 100 mL water and 10 mL of nitric acid. Gently boil the sample on the hot plate until fully dissolved. Cool to room temperature. Add 5 mL of 10% ferric nitrate and mix well. Add 4.00 mL of 0.05N silver nitrate and mix well for about 1 min. Titrate with 0.025N ammonium thiocyanate to a light brown color that persists for at least 15 sec. Record the volume of thiocyanate used. Calculate the percent of chloride calculated as strontium chloride as follows:

$$\text{Percent chloride calculated as strontium nitrate} = \frac{(7.93 \times (4 - (V \times F)) \times N)}{W}$$

Where: V = volume of ammonium thiocyanate used, mL
F = silver nitrate titer of ammonium thiocyanate, mL/ mL

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N = normality of silver nitrate solution, mol/L
W = weight of sample, g

The results for the two replicates shall not differ by more than 0.005 wt% SrCl2, if they do differ, repeat the test. Calculate and report the average of the two replicates.

4.4.11.2 Chloride (alternate method). Chlorides shall be determined turbidimetrically after precipitation with silver nitrate.

4.4.11.2.1 Reagents and stock solutions. All materials used shall be reagent grade or as specified below.

a. Water, deionized or distilled, conforming to ASTM D1193 Type I or II
b. Nitric acid (HNO3), concentrated, 67-70 wt%
c. Silver nitrate, approximately 0.2M, acidified to pH 1-3 with nitric acid.
d. Chloride stock, 12.5 mg/L

4.4.11.2.2 Chloride stock solution preparation. Transfer 652 – 662 mg of potassium chloride to a 250 mL volumetric flask and dilute to volume with water. Mix well and transfer a 1.00 mL aliquot of this solution to a 100-mL volumetric flask. Dilute to volume with water and mix well; the solution serves as stock for further preparation of standards.

4.4.11.2.3 Sample preparation. Prepare two (2) replicates as follows. Dissolve 560 ± 5 mg of the strontium peroxide sample in a pre-mixed solution containing approximately 60 mL of water and 10.0 mL nitric acid. Filter the solution into a 100 mL volumetric thru Whatman No. 40 filter paper that has been pre-washed with water. Rinse the original container with several 5-10 mL portions of water and filter into the flask to collect 100 mL in total. Mix well and transfer a 10.0 mL aliquot of this solution to a 50 mL Nessler tube. Fill up to volume with water.

4.4.11.2.4 Standard solutions preparation. Add 30 mL water and 1.0 mL nitric acid to each of three (3) clean 50-mL Nessler tubes. Add 1.00 mL, 2.00 mL, and 3.00 mL aliquots of the chloride stock to the 1st, 2nd, and 3rd tubes, respectively. Make up to volume with water. The standards correspond to, respectively, 0.05, 0.10, and 0.15 wt% chloride as $SrCl_2$.

4.4.11.2.5 Analysis. To each of the five (5) Nessler tubes (two samples and three standards) add 1.0 mL of 0.2M silver nitrate. Mix well and leave in the dark for ten minutes. Examine the relative opacities of the solutions by looking lengthwise through the Nessler tubes against a dark matte background; allow for adequate lighting but avoid bright lights and reflections. For each of the two samples compare its opacity with those of the standards and identify the closest standard that is opaquer than the sample. This represents the upper limit for chloride content. The sample opacity shall not be greater than 0.10 wt% $SrCl_2$ standard.

Note: The two sample solutions may have slightly different opacities, but both shall have the same upper limit. Otherwise, the analysis shall be repeated ensuring all glassware has been thoroughly cleaned. In case of doubt, the primary back-titration method shall be used.

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4.4.12 Granulation. Nest the U.S. Alternate sieves conforming to ASTM E11, properly superimposed with the coarsest on top, on a bottom pan. Place a 100 g portion of sample on the upper sieve. Add 2 metal disks about the size and weight of nickel coins (5.0 ± 0.2 grams, 2 centimeters in diameter and 2 millimeters thick). Cover and shake for five (5) minutes in a mechanical shaker geared to produce 285 to 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts retained on each sieve and bottom pan and calculate to percent passing on the basis of total sample recovered as follows:

$$\text{Percent through designated sieve} = 100 \times \frac{D}{E}$$

Where: D = Sum of weights of sample recovered from sieves below designated sieve and from bottom pan, g
E = Sum of weights of sample recovered from all sieves and bottom pan, g

4.4.13 Workmanship. Take approximately 300 g of the material and spread the composition out on a clean, well-lighted surface. The sample shall be visually examined, free from visible impurities, and foreign material such as dirt, wooden particles, or any other foreign matter. Failure to meet the workmanship requirement during inspection or while executing production processes shall result in lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Strontium peroxide is used in pyrotechnic mixtures, primarily in tracer igniters and to a lesser extent in basic tracer formulations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Provisions for submission/resubmission of first article samples
c. Quantity required and delivery schedules.
d. Grade of material

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e. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) tested

6.3 Definitions.

6.3.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

6.4 The iron and aluminum can also be determined spectrographycally or by atomic absorption. However, in case of dispute the samples should be analyzed in accordance with 4.4.6, 4.4.8 and 4.4.9.

6.5 The relationships of the grades of MIL-S-612B to the previous MIL-S-612A are as follows:

MIL-S-612B	MIL-S-612A
Grade A deleted	Grade A
Grade B	Grade B
Grade C	Grade C

6.6 Potassium permanganate titrant. Potassium permanganate titrant solution should be acquired commercially. Alternately, the titrant solution could be prepared per literature and standardized against ferrous ammonium sulfate or NIST sodium oxalate. Degradation will occur if potassium permanganate solution is sitting in the burette for more than 30 minutes.

6.7 Technical agency. US Army Combat Capability Development Command Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.8 Approval of equivalent test methods and/or instruments. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.7). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.9 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.7) responsible for preparation of this specification.

6.10 Incorporated changes. ECP Y23Q2009 provides documentation for noted changes.

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6.11 Subject term (key word) listing.

Tracer
Oxidizer

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS

Preparing activity:
Army – AR
(Project No. 6810-2025-001)

Review Activity:
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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